A Project Report on

Smart Classroom system

submitted as CIA3 for the Course

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(EC431P)

by

Name	Kavinraj S
Register No	2460629
Class/Section	4BTEC

FACULTY IN-CHARGE: **COL JAI GOVIND P**

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY

CHRIST (DEEMED TO BE UNIVERSITY)

Academic Year: 2025-26

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY

CERTIFICATE

This is to certify that Mr./Ms. **Kavinraj S**, bearing Register No. **2460629**, has satisfactorily completed the CIA3 project work titled "Smart Classroom system" in the course, EC431P- Artificial Intelligence and Machine Learning prescribed by the Department of Electronics AND COMMUNICATION Engineering, CHRIST (Deemed to be University) for the IV semester B.Tech. program during the academic year 2025-26.

Col Jai Govind P	**Dr Inbanila**
Faculty In-charge	**Head of the Department**

Date: 06/03/2026

Place: Bangalore

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Vision

"Excellence and Service"

Mission

"CHRIST (Deemed to be University) is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment."

Core Values

Faith in God
Moral Uprightness
Love of Fellow Beings
Social Responsibility
Pursuit of Excellence

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Department Vision

"To emerge as a centre of academic excellence in the field of Electronics Communication Engineering to address the dynamic needs of the industry, upholding moral values."

Department Mission

Impart in-depth knowledge in Electronics Communication Engineering to achieve academic excellence. Develop an environment of research to meet the demands of evolving technology. Inculcate ethical values to promote team work and leadership qualities befitting societal requirements. Provide adaptability skills for sustaining in the dynamic environment.

Program Educational Objectives (PEOs)

PEO1: Domain Knowledge *- Apply the knowledge of Electronics Communication Engineering to analyse, design and develop solutions for real time engineering problems.*
PEO2: Research Oriented *- Be competent to pursue higher learning and research.*
PEO3: Ethics Teamwork *- Assimilate technical skills with profes- sional ethics.*
PEO4: Life Long Learning *- Be passionate to attain professional ex- cellence through lifelong learning.*

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop solutions for complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems, reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems, components, or processes to meet identified needs with consideration for public health and safety, wholelife cost, netzero carbon, culture, society, and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge, including design of experiments, modeling, analysis,and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, andmodern engineering and IT tools, including prediction and modeling, recognizing their limitations tosolve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy,health, safety, legal framework, culture, and environment.

PO7: Environment and Sustainability: Recognize the need for, and have the preparation and ability to understand the impact of professional engineering solutions in societal and environmental contexts,and demonstrate knowledge of and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; adhere to national and international laws.

PO9: Individual and Collaborative Teamwork: Function effectively as an individual and as a member or leader in diverse, multi-disciplinary teams.

PO10: Communication: Communicate effectively and inclusively within the engineering communityand society at large, such as being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language, and learning differences.

PO11: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making, and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO12: Life-Long Learning: Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1:RF Microwave Domain - Adapt RF Microwave tools to design and develop antennas and components for the emerging wireless communication systems.

PSO2:Signal Processing Domain - Analyse and apply the princi- ples of signal processing to design and develop solutions for electronics and communication systems.

PSO3: VLSI Embedded Systems Domain - Analyse, design and develop electronic systems to solve real world problems in VLSI Embedded Systems.

PSO4:Devices Circuits Domain - Select, specify and test elec- tronic devices used in analog and digital application circuits.

Evaluation Rubrics

- Laboratory Marks (50M) = Lab Experiments (25M) + Lab Project (25M)
- The lab project project is assessed for 100 marks and scaled down to 25 Marks.
- The following evaluation rubrics followed for assessment:
  - Problem formulation: 10 marks
  - Literature survey: 10 marks
  - Project design or methodology: 10 marks
  - Implementation of project: 20 marks
  - Result analysis: 10 marks
  - Project impact: 5 marks
  - Cost estimation: 5 marks
  - Presentation skill: 5 marks
  - Team work: 5 marks
  - Novelty and Originality: 10 marks
  - Project report : 10 marks

No.	Component	Date	Marks	Signature
1	Problem formulation(10)			
	Literature survey(10)			
2	Project design or methodology (10)			
	Implementation of project (20)			
3	Result analysis (10)			
	Project impact (5)			
	Cost estimation (5)			
4	Presentation skill (5)			
	Team work (5)			
	Novelty and Originality (10)			
	Project report (10)			
Total (100)				
Total (25)				

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CHAPTER 1

Introduction

CONTEXT AND BACKGROUND STUDY

With the rapid advancement of technology, educational environments are increasingly integrating smart systems to improve efficiency, automation, and energy management. Traditional classrooms often rely on manual control of electrical appliances such as fans and lights. As a result, these devices frequently remain switched on even when no students are present, leading to unnecessary energy consumption and increased electricity costs.

Recent developments in Artificial Intelligence (AI), Internet of Things (IoT), and computer vision technologies have enabled the creation of intelligent automation systems capable of monitoring environments and making real-time decisions. Computer vision techniques allow systems to detect and analyze objects in images and video streams. One of the most widely used object detection algorithms is YOLO (You Only Look Once), which is known for its high detection speed and accuracy.

In this project, an AI-based Smart Classroom Automation System is developed using a webcam, YOLO object detection, and an ESP32 microcontroller. The system detects students in real time and automatically controls classroom appliances based on student presence and position. The integration of AI and IoT technologies enables efficient classroom management while reducing energy wastage.

NEED ANALYSIS

Energy conservation and efficient resource management are major concerns in educational institutions. In many classrooms, electrical devices such as fans and lights continue to operate even when the room is empty. This occurs because the appliances depend entirely on manual operation by teachers or students.

Traditional automation methods such as motion sensors or timers provide limited solutions. Motion sensors can detect movement but cannot accurately determine the number of people present or their positions inside the classroom. Timers cannot adapt to real-time conditions and often result in inefficient energy use.

Therefore, there is a strong need for an intelligent system that can automatically detect the presence of students and control classroom appliances accordingly. By using computer vision techniques and AI-based detection algorithms, it becomes possible to accurately identify students and control devices in real time. Such a system can significantly reduce energy wastage while providing a more convenient and automated classroom environment.

IDENTIFICATION OF GAPS

Existing classroom automation systems suffer from several limitations. Many currently available systems rely on simple sensors such as Passive Infrared (PIR) sensors to detect motion. Although PIR sensors can identify movement, they cannot determine the number of individuals present or their exact positions within the room.

Another limitation is that most traditional systems operate on basic ON/OFF automation without intelligent decision-making capabilities. These systems are unable to adapt dynamically to changing classroom conditions. In addition, existing systems often lack integration with modern AI technologies that can provide more accurate detection and analysis.

Furthermore, most automation systems control all appliances simultaneously without considering the seating arrangement of students. This means that fans and lights may operate in areas where no students are present, resulting in inefficient energy usage.

The proposed system addresses these gaps by utilizing an AI-based object detection model to detect students and determine their positions within the classroom. By dividing the classroom into left and right sections, the system can independently control appliances in different areas, thereby improving energy efficiency and automation.

PROBLEM STATEMENT

Classrooms in educational institutions frequently experience inefficient energy usage due to manual control of electrical appliances. Fans and lights often remain switched on even when classrooms are empty or partially occupied. Traditional sensor-based automation systems are unable to accurately detect the number of students or their positions inside the classroom.

As a result, there is a need to develop an intelligent automation system capable of detecting student presence and controlling classroom appliances automatically. The proposed solution is an AI-based Smart Classroom Automation System that uses computer vision techniques to detect students in real time and control fans and lights accordingly.

OBJECTIVES

The primary objective of this project is to design and implement a smart classroom automation system using Artificial Intelligence and IoT technologies.

The specific objectives of the project are as follows:

- To develop a computer vision system capable of detecting students using the YOLO object detection algorithm.
- To process real-time video input from a webcam and identify student positions within the classroom.
- To count the number of students on different sides of the classroom.
- To control classroom appliances such as fans and lights automatically based on student presence.
- To implement communication between the AI detection system and the ESP32 microcontroller.

- To provide both automatic and manual control modes for appliance management.
- To reduce energy consumption and improve efficiency in classroom environments.

CHAPTER 2

Execution

ABSTRACT

The abstract represents the working architecture of the proposed AI-based Smart Classroom Automation System. The system uses a webcam to capture real-time classroom video. The video frames are processed on a laptop using the YOLO object detection model to identify students. Based on the detected positions, the system counts students on the left and right sides of the classroom. This information is transmitted to the ESP32 microcontroller through serial communication. The ESP32 then controls classroom appliances such as fans and lights accordingly.

METHODOLOGY FOR THE STUDY

The methodology followed in this project includes several stages starting from video acquisition to device control.

First, a webcam captures live video of the classroom environment. The captured frames are processed using the YOLO object detection algorithm running on the laptop. The algorithm detects students in real-time and determines their positions within the frame.

After detection, the system calculates the number of students present on the left and right sides of the classroom. The processed data is then sent to the ESP32 microcontroller through serial communication. The ESP32 analyzes the received data and controls the fans and lights based on student presence. In addition to automatic operation, the system also supports manual control through a web interface.

MODELING, ANALYSIS & DESIGN

The proposed system consists of two major modules: the AI detection module and the control module.

The AI detection module runs on the laptop and performs real-time human detection using the YOLO deep learning model. The webcam captures classroom images which are processed using OpenCV and YOLO to detect students and determine their positions.

The control module is implemented using the ESP32 microcontroller. It receives the processed data from the laptop through serial communication and controls relay modules connected to fans and lights. The system architecture ensures efficient communication between the AI detection unit and the hardware control unit.

ALGORITHMS AND FLOWCHARTS

The algorithm used in the system is described as follows:

1. Start the system
2. Capture video frame from the webcam
3. Run YOLO object detection
4. Detect and count students
5. Identify left and right student positions
6. Send detection data to ESP32
7. ESP32 processes the received data
8. Control fans and lights accordingly
9. Repeat the process continuously

```
Start
 ↓
Capture Video Frame
 ↓
Run YOLO Detection
 ↓
Count Left / Right Students
 ↓
Send Data to ESP32
 ↓
Control Appliances
 ↓
Repeat
```

Figure 2.1: Flowchart

PROTOTYPE & TESTING

A prototype of the smart classroom automation system was developed using an ESP32 microcontroller, relay modules, a webcam, and AI-based detection software. The prototype was tested in a classroom-like environment to evaluate the performance of the system.

Several tests were conducted including camera detection accuracy, student counting reliability, response time of the system, and proper functioning of fan and light control. The system successfully detected students and automatically controlled appliances based on their positions.

The testing results demonstrated that the proposed system can effectively reduce unnecessary energy consumption while maintaining a comfortable classroom environment.

CHAPTER 3

Outcomes

RESULTS & ANALYSIS

The proposed AI-based Smart Classroom Automation System was successfully implemented and tested using a prototype setup consisting of a laptop running the YOLO object detection model, an ESP32 microcontroller, two DC motors representing fans, a light LED representing classroom lighting, and a DHT temperature sensor.

The webcam captures real-time video frames of the classroom environment. These frames are processed using the YOLO object detection algorithm running on the laptop. The system detects the presence of students and determines whether they are located on the left or right side of the classroom.

Based on the detection results, the system sends control signals to the ESP32 microcontroller through serial communication. The ESP32 then activates the corresponding fans and lights according to student presence. Figure 3.1 shows the hardware prototype of the system where two DC motors represent the left and right fans, and an LED represents the classroom light.

Figure 3.1: Hardware Setup of Smart Classroom System

Figure **??** shows the working demonstration of the system. The YOLO detection model identifies a person in the camera frame and determines the position of the student relative to the center line. Based on the detected position, the system activates the corresponding fan and lighting system.
The system was tested under different scenarios to verify its functionality.

Table 3.1: System Operation Based on Student Presence

Students Detected	Light	Left Fan	Right Fan
No Students	OFF	OFF	OFF
Left Side Student	ON	ON	OFF
Right Side Student	ON	OFF	ON
Students on Both Sides	ON	ON	ON

The experimental results show that the system successfully detects student positions and controls classroom appliances automatically. This helps in reducing unnecessary energy consumption when classrooms are empty.

COMPARATIVE STUDY

A comparison between the traditional classroom system and the proposed smart classroom automation system is shown below.

Feature	Traditional System	Proposed System
Automation	Manual Control	Automatic Control
Student Detection	Not Available	YOLO AI Detection
Energy Efficiency	Low	High
Position Detection	Not Available	Left / Right Detection
Smart Control	No	Yes

DISCUSSIONS

The proposed smart classroom system demonstrates how artificial intelligence and IoT technologies can be integrated to automate classroom appliances. The system effectively detects human presence using computer vision and controls fans and lights accordingly.
One of the major advantages of this system is improved energy efficiency, as appliances are automatically turned off when no students are detected. However, detection accuracy may slightly reduce under poor lighting conditions or if the camera view is obstructed.
Despite these minor limitations, the system proves to be an effective solution for intelligent classroom automation.

Component	Approximate Cost (INR)
ESP32 Microcontroller	350
Relay Module	120
DHT22 Temperature Sensor	200
Webcam	600
DC Motors (Fans Model)	150
Miscellaneous Components	200
Total Cost	1620

PROJECT COST ESTIMATION

PROJECT IMPACT

3.5.1 Environmental Impact

The proposed system helps reduce unnecessary electricity consumption by automatically switching off classroom appliances when no students are present. This contributes to energy conservation and promotes sustainable usage of electrical resources.

3.5.2 Societal Impact

Smart classroom technologies enhance the learning environment by introducing modern automation systems in educational institutions. It also helps students understand the practical applications of artificial intelligence and IoT.

3.5.3 Economical Impact

By reducing energy wastage, the system lowers electricity costs for educational institutions and contributes to efficient resource management.

SDG COMPLIANCE

The project supports the following United Nations Sustainable Development Goals:

- SDG 7: Affordable and Clean Energy
- SDG 9: Industry, Innovation and Infrastructure
- SDG 11: Sustainable Cities and Communities

CONCLUSIONS

The AI-based Smart Classroom Automation System successfully demonstrates the integration of computer vision and IoT technologies to create an intelligent classroom environment. The system detects students using the YOLO object detection algorithm and controls fans and lights based on their positions in the classroom.
The implementation of this system helps reduce energy wastage, improves automation, and provides a modern smart classroom solution.

SCOPE FOR FUTURE WORK

Future improvements to the system may include:

- Automatic attendance system using face recognition
- Cloud-based monitoring and data analytics
- Mobile application control for teachers
- Seat-level student detection
- Smart fan speed control based on temperature

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